Mode

- Single-omics
- Multi-omics

Alzheimer Agent — Single-omics data processing and analysis

Agent plan Harmonize PatternMine Hypothesize KBPrune ProgressModel status: live

› Agent action log

Drop your files

Drag and drop files here
Limit 4GB per file • CSV
Browse files

GSE138852_counts.csv 273.9MB

GSE138852_covariates.csv 0.8MB

Output folder

outputs

GSE138852_covariates.csvGSE138852_counts.csv

Type a note… Shift+Enter = newline; Enter = Send

Send

Plan: I'll first harmonize the counts (QC → normalize/log1p → RAW snapshot) , then mine patterns (HVGs → PCA → neighbors/UMAP → Leiden + markers) , then test AD vs normal and summarize hypotheses.

- **Harmonize** — QC advisor; normalize/log1p; RAW snapshot.
- **PatternMine** — HVGs/PCA → neighbors/UMAP → Leiden + markers.
- **Hypothesize** — AD vs normal DE, ranking and outputs.

A) Load counts → AnnData ✓

⌄ A) Load counts → AnnData

Cells (before filter): **13214**, Genes: **10850**

- Matrix transposed to cells×genes; raw counts stored for later.
- Proceeding to QC metric computation.

B) QC metrics & plots ✅

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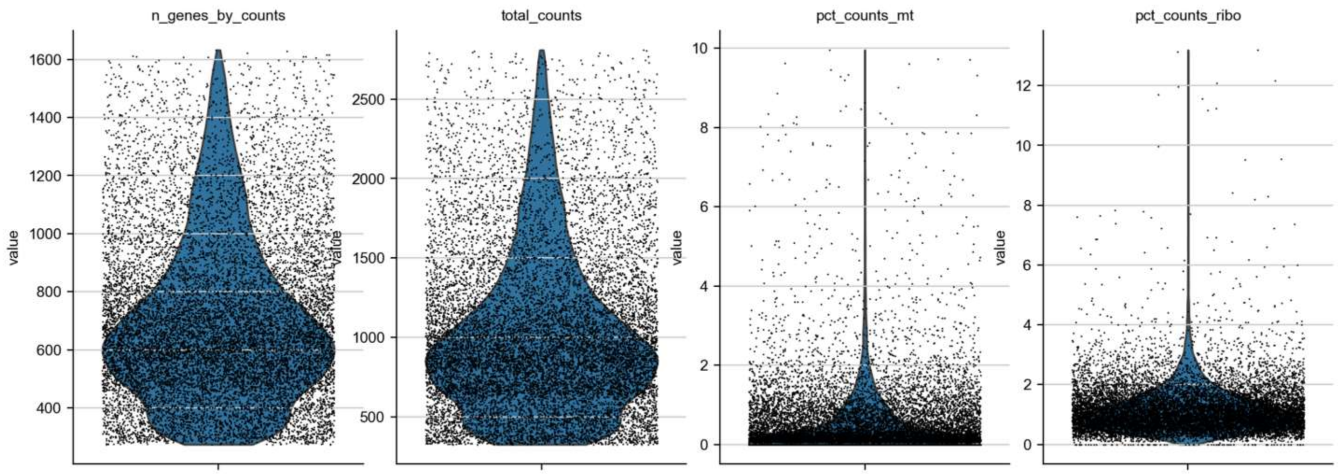

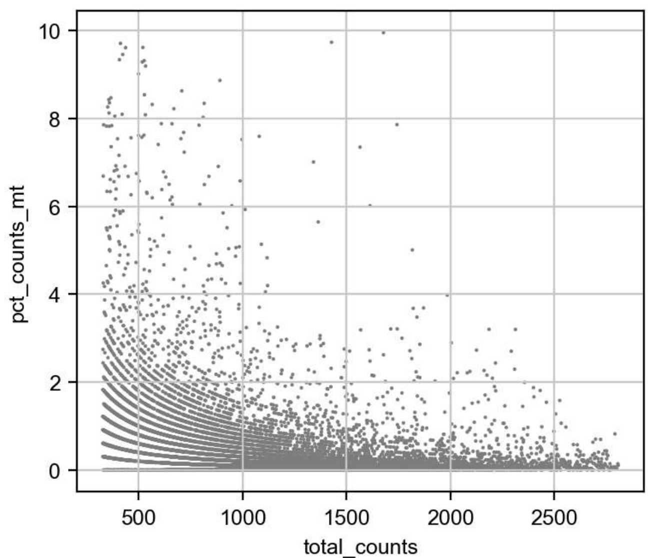

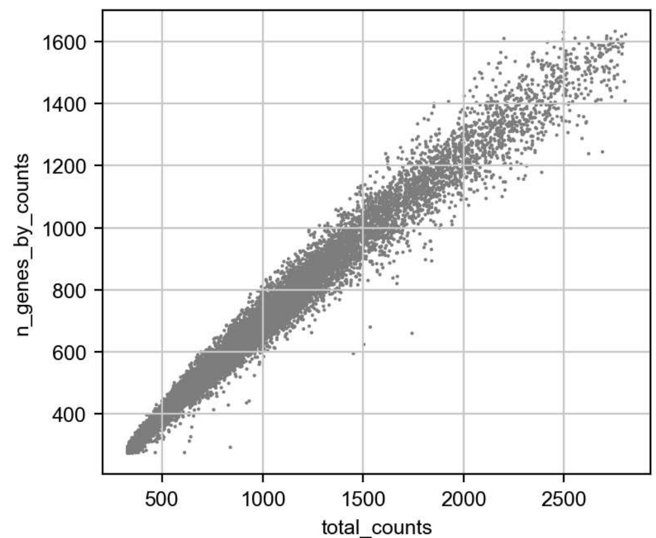

Auto-QC Advisor

- `n_genes_by_counts` — values top out around **1,632**. 👉 The 99–99.5th pct suggests a cap near **1,550**.

C) Filtering ✅

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After filtering: **13036** cells × **10850** genes (before: 13214 × 10850)

Decision

- Based on QC distributions, I used **mt% < 5%** and **n_genes_by_counts < 1550**.
- This keeps the dense core while trimming long tails.

D) Normalize / log1p / RAW snapshot ✅

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Saved normalized matrix → normalized_log1p.h5ad

Decision

- Per-cell library size normalization (1e4) followed by log1p; raw counts kept in `layers['counts']`.
- Snapshot saved for downstream reuse.

E) HVGs → Scale → PCA ✅

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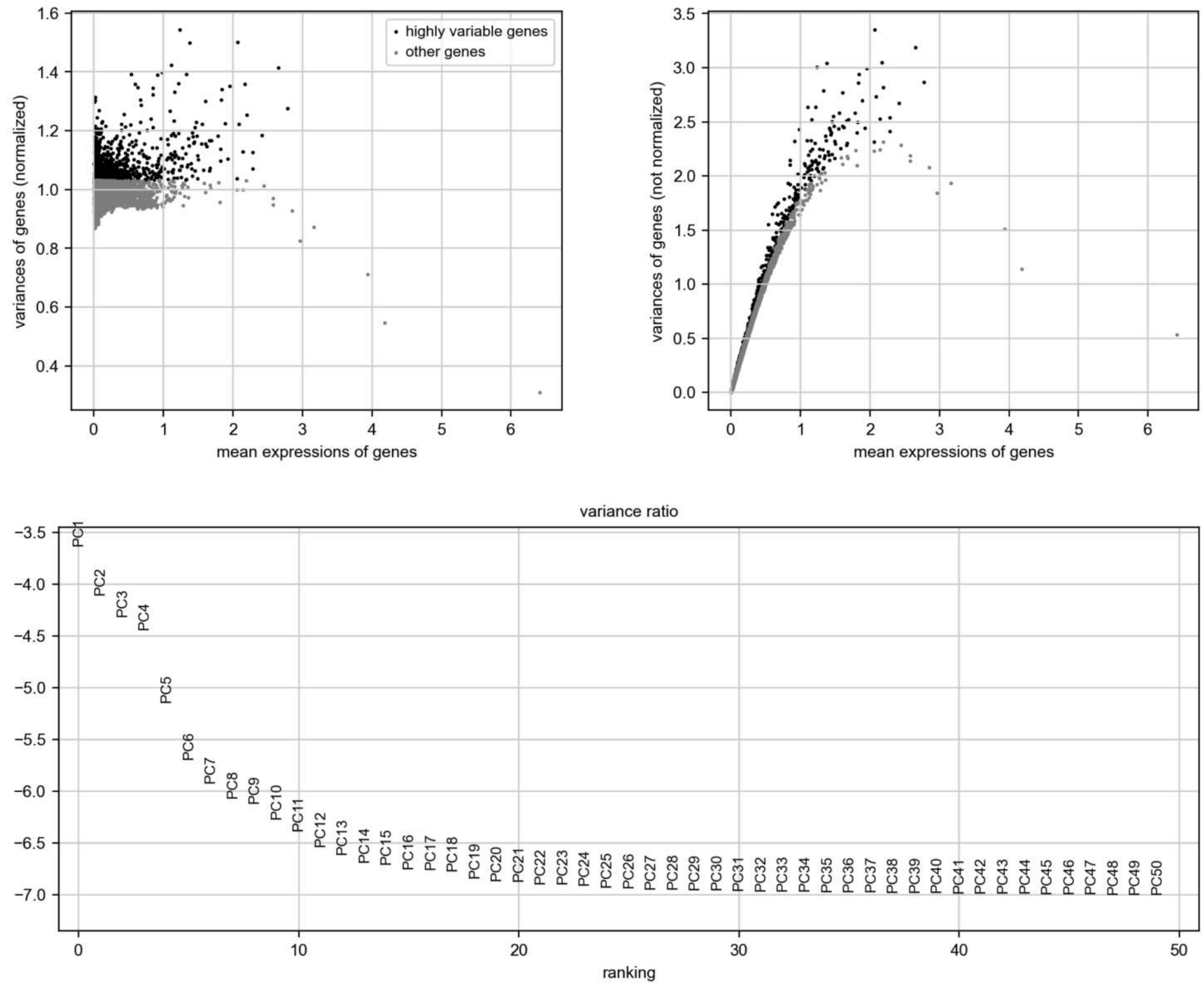
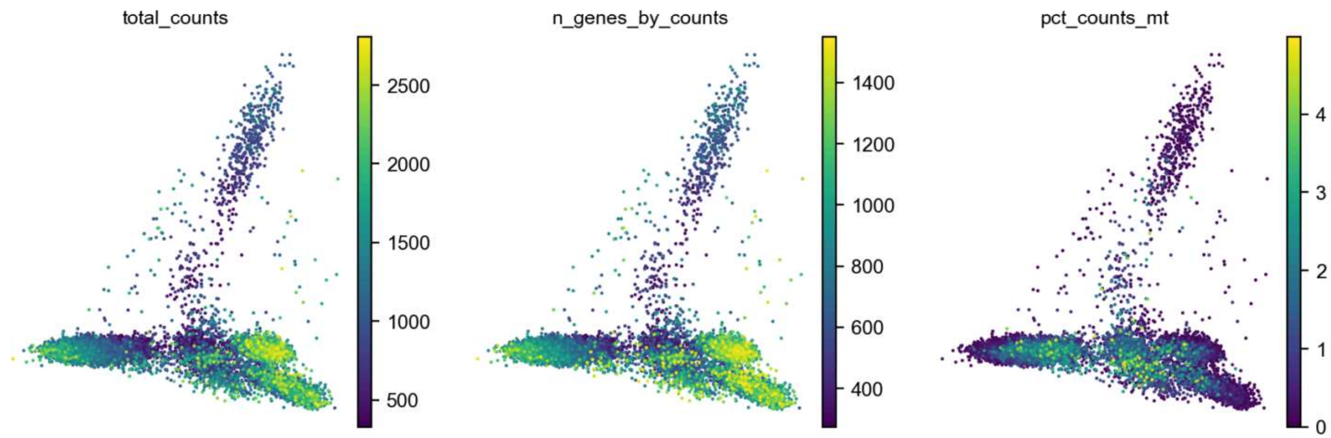


highly variable genes
other genes
variances of genes (normalized)
mean expressions of genes
variances of genes (not normalized)
mean expressions of genes
variance ratio
ranking



Decision

- Picked **2000 HVGs** (Seurat v3).
- PCs explain 90% variance by about **PC1**; we'll keep 50 for neighbors/UMAP.

F) Neighbors → UMAP ✅

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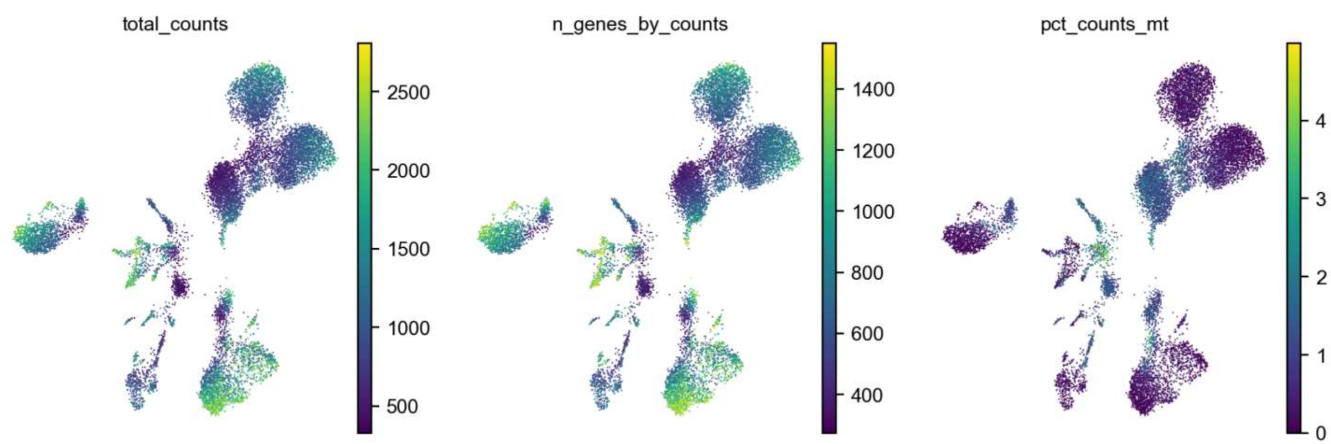

Decision

- **n_neighbors=20** as a balanced default for datasets of this scale.
- Fixed random_state for reproducible UMAP layout.

G) Join covariates & color UMAP ✅

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H) Leiden clustering & markers ✅

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I) Differential expression — AD vs normal ✅

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J) Enrichment of AD-up signature ✅

⌄ J) Enrichment of AD-up signature

	Rank	Term	Overlap
0	1	response to unfolded protein (GO:0006986)	9/49
1	2	regulation of cellular response to stress (GO:0080135)	11/118
2	3	regulation of cellular response to heat (GO:1900034)	9/79
3	4	regulation of apoptotic process (GO:0042981)	24/742
4	5	regulation of protein modification by small protein conjugation or removal (GO:1903320)	5/20
5	6	negative regulation of inclusion body assembly (GO:0090084)	4/10
6	7	regulation of inclusion body assembly (GO:0090083)	4/11
7	8	negative regulation of axonogenesis (GO:0050771)	5/28
8	9	myelination (GO:0042552)	6/48
9	10	Protein processing in endoplasmic reticulum	10/171

Download enrichment (AD-up)

K) Hypotheses & follow-ups ✅

⌄ K) Hypotheses & follow-ups

Takeaways:

- AD cells show proteostasis / ER-stress (UPR) and heat-shock responses.
- Signals in apoptosis / protein-aggregation pathways.

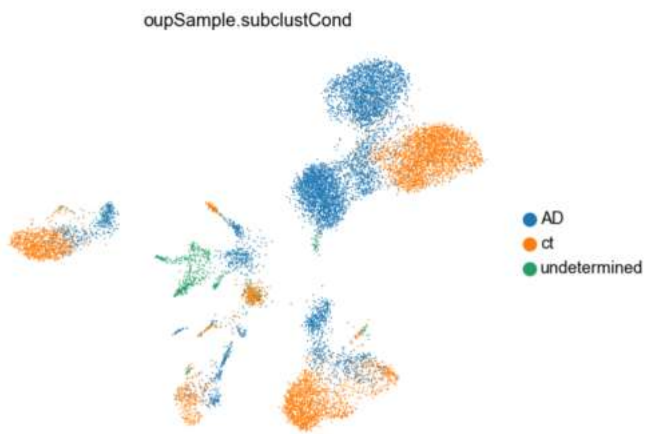
Suggested follow-ups:

- Run enrichment on AD-down genes as well.
- Repeat within major cell types to localize signals.
- Plot key markers (e.g., *HSPA1A*/*HSP90*, UPR genes, myelin genes) on UMAP/violin.

Done.

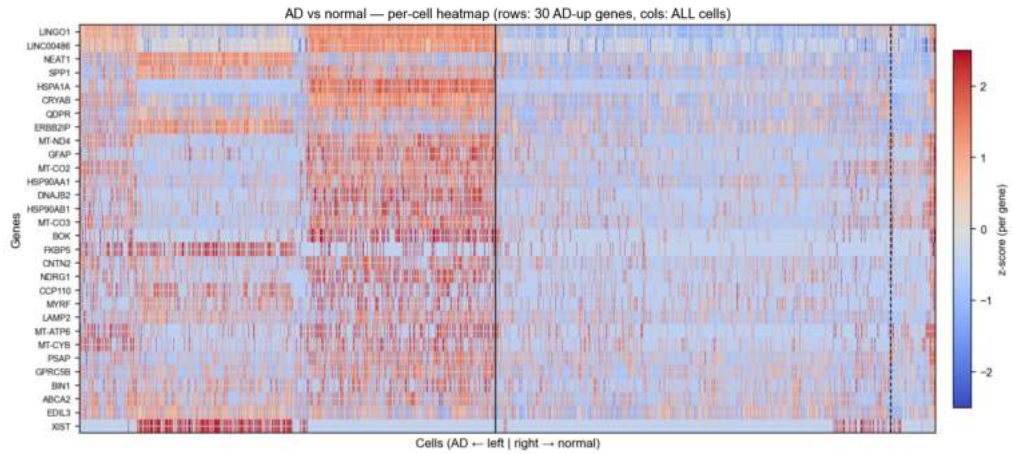
Results summary

Here's the supervised UMAP colored by condition (AD vs *normal*).



`None`

This is the per-cell heatmap of the top 30 AD-up genes across all cells (AD left, normal right).



`None`

Harmonized (normalized) matrix saved: `normalized_log1p.h5ad` in `outputs`.

Candidate genes

AD-up (top 10):

`LINGO1`, `LINC00486`, `NEAT1`, `SPP1`, `HSPA1A`, `CRYAB`, `QDPR`, `ERBB2IP`, `MT-ND4`, `GFAP`

AD-down (top 10):

`NTM`, `LSAMP`, `NRXN1`, `CTNNA2`, `LRP1B`, `GPM6A`, `RORA`, `CTNND2`, `DPP10`, `SLC1A2`

Hypotheses & follow-ups

Based on the Differential expression of AD vs normal and adjusted p value, **LINGO1** and **LINC00486** are likely to be related with ADRD.

Based on the Enrichment of AD-up signature:

- AD cells show proteostasis/ER stress (UPR) and heat-shock responses.
- There's signal in apoptosis/protein aggregation pathways.
- There may be myelin/axon remodeling involvement.

Suggested follow-ups:

- Run enrichment on AD-down genes too.
- Repeat within major cell types.
- Check key marker genes (e.g., HSPA1A/HSP90, UPR genes, myelin genes) on UMAP/violin.

Saved QC and Results

Download — QC scatter mt% vs total

Download — QC scatter n_genes vs total

Download — PCA variance ratio

Download — UMAP (unsupervised panel)

Download — UMAP (condition AD vs normal)

Download — Heatmap (AD-up 30 genes)

Download — DE table (AD vs normal)

Download — Normalized matrix

Agent suggestions

Try a next step — clicking a chip will pre-fill the message box in the sticky footer.

Run cell-type DE | Tune QC cutoffs | Explain cluster 7

Why these suggestions?